

P1. $G = H = \mathbb{Z}_2$. Then $G \times H$ contains no element of order 4.

P2. A group of order 6 is either $\mathbb{Z}_6$ or $\mathbb{Z}_2 \times \mathbb{Z}_3$. A_4 has no element of order 6.

Hence, if such a group exists, it must be isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_3$, which requires

a 3-cycle and $(12)(34)$ commute, WLOG. Equivalently, a 3-cycle

is conjugate to itself after operating by $(12)(34)$.

This is impossible by checking $(123) \longrightarrow (214)$, etc.

P3. (a) 3, 3, 4; 4, 3; 3.

(b) $d_{\min} = 3$.

(c) $d_{\min} - 1 = 2$.

(d) $00111 \xrightarrow{\text{mistake}} 10111$ detected.

(e) $00000 \xrightarrow{\text{mistake}} 11100$ not detected.

P4 (a) $f(1011) = 10111$; $f(0100) = 01001$.

(b) If odd number of bits flip, then the parity of # "1" is changed.

(c) 00000 and 11000 have the same last-digit, differed by two digits, and are both 'legal'.